

ENTHUSIAST ADVANCE COURSE

PHASE : MEA

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 23-04-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	1	3	3	2	2	3	2	4	1	1	4	1	1	3	3	2	1	1	4	4	1	2	3	4	3	2	1	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	3	4	1	3	1	2	2	3	3	1	3	3	1	1	1	3	4	1	4	1	3	4	4	3	4	4	3	4	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	4	2	3	1	1	4	1	4	2	1	3	1	1	2	3	1	2	4	3	4	3	2	4	1	4	4	3	1	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	4	3	4	1	1	1	4	1	3	4	2	4	2	4	3	4	1	1	2	4	2	3	1	1	3	4	3	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	2	2	4	4	3	3	4	3	2	4	1	4	2	2	1	2	3	4	1	3	3	4	3	3	4	1	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	3	4	4	2	4	1	4	1	1	2	3	4	2	2	2	3	4	4	4	3	3	1	3	3	3	4	2	3	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	2	4	1	2	1	2	4	1	2	4	2	3	3	2	4	1	2	3	2										

HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**1. **Ans (2)**

$$u = -200 \text{ cm}$$

$$v = -100 \text{ cm}$$

$$f = ?$$

$$\frac{1}{-100} - \frac{1}{-200} = \frac{1}{f} \Rightarrow \frac{1}{f} = -\frac{1}{200}$$

$$P = \frac{100}{f(\text{cm})} = -\frac{1}{2} = -0.5 \text{ D}$$

2. **Ans (1)**

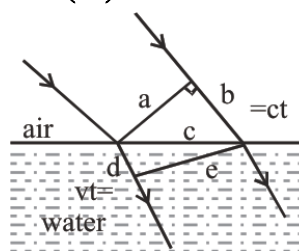
$$\text{Magnification } M = \frac{V_0}{u_0} \left(1 + \frac{D}{f_c} \right)$$

$$\text{for objective } \frac{1}{V_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\Rightarrow \frac{1}{V_0} - \frac{1}{(-5)} = \frac{1}{4}$$

$$\Rightarrow V_0 = 20 \text{ cm}$$

$$M = \frac{20}{5} \left(1 + \frac{20}{10} \right) \Rightarrow M = 12$$

3. **Ans (3)**

Time taken by wave in air to travel distance

$$t = \frac{b}{c}$$

During same time distance travelled in medium.

$$d = tv \quad \text{so} \quad \mu = \frac{c}{v} = \frac{b/t}{d/t} = \frac{b}{d}$$

4. **Ans (3)**

$$w = \frac{\lambda D}{d} \Rightarrow w_{\text{net}} = \frac{\lambda(2D)}{d/2} = \frac{4\lambda D}{d} = 4w$$

5. **Ans (2)**

Resultant intensity $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$

At central position with coherent source (and $(I_1 = I_2 = I_0)$.

$$I_{\text{coh}} = 4I_0 \quad \dots(i)$$

In case of incoherent at a given point, ϕ varies randomly with time so $(\cos \phi)_{\text{av}} = 0$.

$$\therefore I_{\text{Incoh}} = I_1 + I_2 = 2I_0 \quad \dots(ii)$$

$$\text{Hence } \frac{I_{\text{coh}}}{I_{\text{Incoh}}} = \frac{2}{1}.$$

6. **Ans (2)**

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$$

$$I = 8 \cos^2 \left(\frac{\pi}{6} \right) = 6 \text{ mW/m}^2$$

7. **Ans (3)**

$$y_1 = \frac{n_1 \lambda_1 D}{d} \text{ for bright fringes}$$

$$y_2 = \frac{n_2 \lambda_2 D}{d} \text{ for bright fringes}$$

To coincide

$$n_1 \lambda_1 = n_2 \lambda_2$$

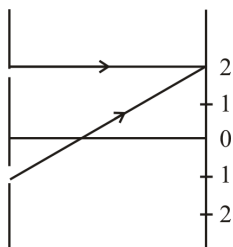
$$n_1 \times 650 = n_2 \times 520$$

$$\frac{n_1}{n_2} = \frac{4}{5}$$

For minimum value of n_1 and n_2

$$n_1 = 4 \text{ \& } n_2 = 5$$

$$(y_1)_{\text{min}} = 4 \times 650 \times 10^{-9} \text{ m} \times \frac{150 \times 10^{-2}}{0.5 \times 10^{-3} \text{ m}} = 7800 \times 10^{-6} \text{ m} = 7.8 \text{ mm}$$

8. **Ans (2)**

for maxima path difference $= n\lambda$

$$n = 0, \pm 1, \pm 2 \dots$$

(Here we will get 5 maxima (one central maxima, two maxima on right side and two maxima on left side.)

9. **Ans (4)**

$$\beta = \frac{\lambda D}{d} \Rightarrow d = \frac{\lambda D}{\beta}$$

$$= \frac{6000 \times 10^{-10} \times (40 \times 10^{-2})}{0.012 \times 10^{-2}} = 0.2 \text{ cm}$$

10. **Ans (1)**

$$\frac{9D\lambda}{d} - \frac{3}{2} \frac{D\lambda}{d} = 7.5 \text{ mm}, d = 0.5 \times 10^{-6}$$

$$\frac{D\lambda}{d} [9 - 1.5] = 7.5 \text{ mm}, d = 5 \times 10^{-7}$$

$$\frac{D\lambda}{d} = 1 \text{ mm}$$

$$\frac{1 \times \lambda}{0.5 \times 10^{-3}} = 1 \times 10^{-3}$$

$$\Rightarrow \lambda = 5000 \text{ \AA}$$

11. **Ans (1)**

$$(\mu - 1)t = n\lambda$$

$$n = \frac{(1.5 - 1) \times 4 \times 10^{-6}}{5 \times 10^{-7}}$$

$$n = 4$$

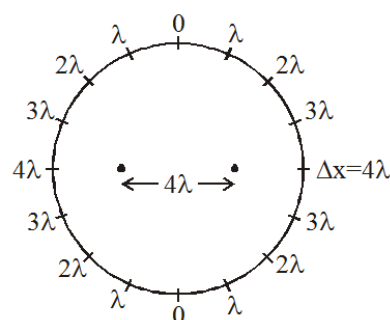
12. **Ans (4)**

At central bright fringe, there is zero order

maxima of all wavelength.

But higher order maxima are at different

positions for different wavelength

13. **Ans (1)**

No. of maxima = 16.

14. **Ans (1)**

$$\text{Path difference } \Delta x = \frac{yd}{D}$$

$$\text{Here } y = \frac{d}{2} = \frac{5\lambda}{2} \quad (\text{as } d = 5\lambda)$$

$$\text{and } D = 10d = 50\lambda$$

$$\text{So, } \Delta x = \frac{5\lambda}{2} \times \frac{5\lambda}{50\lambda} = \frac{\lambda}{4}$$

Corresponding phase difference will be

$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x = \left(\frac{2\pi}{\lambda} \right) \left(\frac{\lambda}{4} \right)$$

$$= \frac{\pi}{2}$$

$$\text{or } \frac{\Delta\phi}{2} = \frac{\pi}{4}$$

$$\begin{aligned} \therefore I &= I_0 \cos^2 \left(\frac{\Delta\phi}{2} \right) \\ &= I_0 \cos^2 \left(\frac{\pi}{4} \right) = \frac{I_0}{2} \end{aligned}$$

15. **Ans (3)**Rarer to denser reflection phase shift = π .16. **Ans (3)**

Thin film interference.

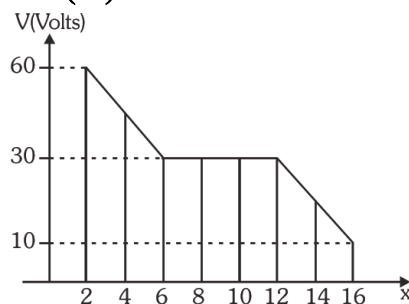
17. **Ans (2)**Center is bright, when phase difference $2n\pi$ or path difference $n\lambda$ 19. **Ans (1)**

In the direction of electric field, potential decreases

23. **Ans (2)**

The work done required is equal to the potential energy of the system. Thus,

$$\begin{aligned} W = U &= \frac{1}{4\pi\epsilon_0} \left[\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_3q_4}{r_{34}} + \frac{q_4q_1}{r_{41}} + \frac{q_1q_3}{r_{13}} + \frac{q_2q_4}{r_{24}} \right] \\ &= \frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{a} - \frac{q^2}{a} - \frac{q^2}{a} - \frac{q^2}{a} + \frac{q^2}{a\sqrt{2}} + \frac{q^2}{a\sqrt{2}} \right] \\ &= \frac{1}{4\pi\epsilon_0} \left[-\frac{4q^2}{a} + \frac{2q^2}{a\sqrt{2}} \right] \\ &= -\frac{q^2}{4\pi\epsilon_0 a} [4 - \sqrt{2}] = -\frac{2.6}{4\pi\epsilon_0} \left(\frac{q^2}{a} \right) \end{aligned}$$

24. **Ans (3)**

$$E = -\frac{dV}{dx} \quad (\text{-ve slope of V-x curve})$$

electric field at 13m is

$$= \frac{30 - 10}{16 - 12} = \frac{20}{4} = 5 \text{ (V/m)}$$

26. **Ans (3)**

$$W = PE (\cos\theta_1 - \cos\theta_2)$$

$$W = PE (\cos 0^\circ - \cos 60^\circ)$$

$$= PE \left(1 - \frac{1}{2} \right) = \frac{PE}{2}$$

$$W' = PE (\cos 0^\circ - \cos 180^\circ) = 2PE$$

$$= 4W$$

30. **Ans (1)**

$$\frac{E}{\rho} = J \Rightarrow E = J\rho \Rightarrow E = \frac{JAR}{\ell}$$

SECTION-B

36. **Ans (1)**

$$MP = \frac{400}{10} = 40$$

$$\frac{\beta \times 3.8 \times 10^5}{3.5 \times 10^3} = 40$$

$$\beta = \frac{40 \times 3.5 \times 10^3}{3.8 \times 10^5}$$

$$\beta = 0.36 \text{ radian}$$

37. **Ans (2)**

$$\frac{I_1}{I_2} = \frac{1}{4}, I_{\max} = 9, I_{\min} = 1$$

$$\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{9 - 1}{9 + 1} = \frac{8}{10} = \frac{4}{5} = 0.8$$

38. **Ans (2)**

$$\theta_a = \theta_w \times \frac{4}{3} = 0.2 \times \frac{4}{3} = 0.27$$

39. **Ans (3)**

By theory

40. **Ans (3)**

For constructive interference in reflected system,

$$2\mu t = (2n+1) \frac{\lambda}{2}$$

$$2 \times 1.4 \times 300 \times 10^{-9} = (2n+1) \frac{\lambda}{2}$$

$$\lambda = \frac{16800}{(2n+1)} \times 10^{-10} \text{ m}$$

$$\text{for } n=0 \quad \lambda_1 = 16800 \text{ \AA}$$

$$\text{for } n=1 \quad \lambda_2 = \frac{16800}{3} \text{ \AA} = 5600 \text{ \AA}$$

$$\text{for } n=2 \quad \lambda_3 = \frac{16800}{5} \text{ \AA} = 3360 \text{ \AA}$$

41. **Ans (1)**

$$\theta = \frac{\Delta x}{a} = \frac{(2n+1)\lambda}{2a}$$

42. **Ans (3)**

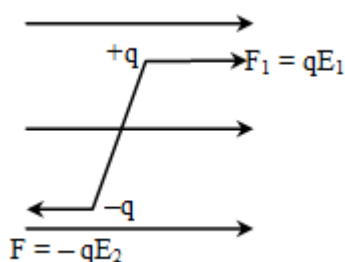
$$\frac{7\lambda D}{2a} = \frac{5\lambda_{\text{red}} D}{2a}$$

$$= 7\lambda = 5 \times 6500 \text{ \AA}$$

$$\lambda = \frac{5 \times 6500}{7} = 4642.8 \text{ \AA}$$

43. **Ans (3)**

Assertion is correct but reason is false, because as the waves diffracted at the edges of circular obstacle, placed in the path of light interfere constructively at the centre of the shadow resulting in formation of bright spot.

46. **Ans (1)**

In non-uniform field both force and torque is non zero.

SUBJECT : CHEMISTRY**SECTION-A**58. **Ans (3)**

In vapour phase of sulphur S_2 molecule has 2 unpaired

61. **Ans (1)**(c) $Al(OH)_3 \downarrow$ ppt. Highly insoluble70. **Ans (2)**

$$\text{Reducing power} \propto \frac{1}{SRP(E^\circ)}$$

71. **Ans (1)**

$$\Lambda_m = \frac{10^{-3} \times K}{M}$$

$$\Lambda_m = \frac{10^{-3} \times 9.2 \times 10^{-2}}{0.02}$$

$$\Lambda_m = \frac{9.2 \times 10^{-5} \times 100}{2}$$

$$= 4.6 \times 10^{-3}$$

72. **Ans (3)**

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.059}{2} \log \frac{[Sn^{2+}]}{[Ag^+]^2}$$

Ag^+ increase, E_{cell} increase.

74. **Ans (1)**

$$\Delta G^\circ = -nFE_{\text{Cell}}^\circ \Rightarrow E_{\text{Cell}}^\circ = \frac{-96.5 \times 1000}{-96500 \times 2}$$

$$= +0.5 \text{ volt}$$

This is oxidation ($M \rightarrow M^{+2}$) potential

$\therefore SRP = +0.5 \text{ volt}$

75. **Ans (2)**

$$\text{eq.(1)} - \text{eq.(2)} = \text{eq.(3)}$$

$$E_3^\circ = \frac{n_1 E_1^\circ - n_2 E_2^\circ}{n_3}$$

$$= \frac{3[-0.74] - [-0.40]}{2} = -0.91 \text{ V}$$

76. **Ans (3)**

$$\alpha = \frac{\Lambda_{\text{eq}}^c}{\Lambda_{\text{eq}}^\infty} = \frac{10}{200} = 0.05$$

77. **Ans (1)**

$$E^\circ = \frac{0.059}{n} \log k_{\text{eq}}$$

$$0.59 = \frac{0.059}{1} \log k_{\text{eq}}$$

$$\log k_{\text{eq}} = 10$$

$$k_{\text{eq}} = 10^{10}$$

78. **Ans (2)**

NCERT Pg.No. # 85

113. **Ans (3)**

NCERT Pg. # 251

114. **Ans (1)**

NCERT Pg.# 250

117. **Ans (4)**

NCERT Pg. # 236, 237, 238

120. **Ans (2)**

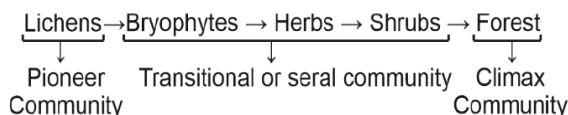
NCERT-XII Pg. # 250

122. **Ans (4)**

NCERT Pg. # 251

123. **Ans (2)**

Sequence of various stages in xerarch succession is as:

124. **Ans (2)**

NCERT Pg. # 252

125. **Ans (4)**

NCERT XII, Pg. No. 263

129. **Ans (4)**

NCERT Pg. # 232

131. **Ans (2)**NCERT (XIIth) Pg. # 250, 251135. **Ans (2)**

NCERT Pg. # 237

SECTION-B138. **Ans (2)**

NCERT-XII Pg. # 250

139. **Ans (3)**

NCERT Pg. # 220

140. **Ans (4)**

NCERT Pg. # 220

142. **Ans (3)**

NCERT Pg. # 236, 237, 238

145. **Ans (3)**

NCERT Pg. # 220

147. **Ans (4)**

NCERT-XII, Pg. # 221 [E], 240 [H]

150. **Ans (1)**

NCERT Pg. # 232

SUBJECT : BIOLOGY-II**SECTION-A**156. **Ans (4)**NCERT (XIIth) Pg. # 101157. **Ans (1)**

NCERT (XII) Pg. # 101, 102

158. **Ans (4)**

Module-1 Pg # 135

159. **Ans (1)**

NCERT XII Pg, 103,104

160. **Ans (1)**

NCERT Pg. # 100, 101

166. **Ans (2)**

NCERT XII pg.#103 (E), 112 (H)

167. **Ans (3)**

NCERT XII Pg, 100,101,105,106

172. **Ans (3)**

NCERT XII Pg, 106,107

177. **Ans (4)**

NCERT XII Pg, 107

182. **Ans (2)**

NCERT XII Pg, 101

184. **Ans (1)**

NCERT XII Pg, 96

185. **Ans (2)**

NCERT XII Pg, 107

SECTION-B186. **Ans (1)**NCERT (XIth) Pg. # 152187. **Ans (2)**NCERT XIIth Pg.#99(E), 107(H) Figure-6.4(a)

188. Ans (4)

NCERT XII Pg, 103

189. Ans (1)

NCERT XII Pg, 103

191. Ans (4)

NCERT XII Pg, 106

193. Ans (3)

NCERT (XII) Pg. # 69,97

196. Ans (4)

NCERT XII Pg, 97

197. Ans (1)

NCERT XII Pg, 100,101,102

198. Ans (2)

NCERT XII Pg, 104

200. Ans (2)

NCERT XII Pg, 99,101,104,106